

Ratnakirti Roy, Ph.D.

Professional Member, IEEE

30, 3rd Cross, Padmanabha Nagar, Achit Nagara P.O, Bengaluru -560107

Phone: +91-9832346296/ +91- 9434379777

Email: rroy.nitdgp@gmail.com

Linkedin: <https://in.linkedin.com/in/ratnakirtiroy>

Google Scholar: <https://scholar.google.com/citations?user=vKjSu1sAAAAJ&hl=en>

ORCID: <https://orcid.org/0000-0001-6259-7197>

SCOPUS Author ID: **55644570600**

Educational Qualifications

<i>Qualification</i>	<i>Board/University</i>	<i>Year</i>	<i>Percentage</i>	<i>Grade/Div.</i>
Ph. D	National Institute of Technology, Durgapur	2017	-	(Awarded)
Master of Computer Applications (MCA)	University of North Bengal	2008-2011	81.82	First
Bachelor of Computer Applications (BCA)	University of North Bengal	2005-2008	72.78	First
Higher Secondary (Class XII) Stream: Pure Science	West Bengal Council for Higher Secondary Education	2005	88.20	First
Madhyamik Pariksha (Class X - Secondary level)	West Bengal Board of Secondary Education	2003	86.75	First
GATE	Indian Institute of Technology	2011	Percentile: 94%	Qualified

Work Experience

- Associate Professor, Acharya Institute of Technology, Bangalore (December 2023 - Present)
- Associate Professor, Dr. B. C. Roy Engineering College, Durgapur (May 2022 - December 2023)
- Assistant Professor, Dr. B.C. Roy Engineering College, Durgapur (January 2019 – April 2022)
- Assistant Professor, School Of Computer Science, Inspiria Knowledge Campus, Siliguri (April 2015 – December 2018)
- Doctoral Research Scholar (Full Time), National Institute of Technology, Durgapur (April 2012 to May 2016)
- Analytics Executive, Abacus Analytics (Kantar IMRB), Kolkata (September 2011 to March 2012)
- Computer Faculty, North Bengal St. Xavier's College, Rajgunj, Jalpaiguri, West Bengal (August 2011 to September 2011)

Administrative Experience

- i) Head (Dept. of MCA), Acharya Institute of Technology, Bangalore (December 2023 - present)
- ii) Registrar, Dr. B. C. Roy Engineering College, Academy of Professional Courses, Durgapur (May 2022 - December 2023)
- iii) Assistant Registrar, Dr. B. C. Roy Engineering College, Academy of Professional Courses, Durgapur (October 2021 - April 2022)
- iv) Head (Digital Promotions), Dr. B. C. Roy Engineering College, Academy of Professional Courses, Durgapur (February 2021 - October 2021)
- v) Head (Academic Program Coordinator), School of Computer Science, Inspiria Knowledge Campus, Siliguri (November 2015 – December 2018)
- vi) Institutional examination cell member, Inspiria Knowledge Campus, Siliguri (September 2015 – December 2018)
- vii) Institutional grievance redressal cell member, Inspiria Knowledge Campus, Siliguri (September 2015 – December 2018)

Awards and Recognition

1. **University first rank medal** for securing the highest marks in BCA examinations (Session: 2005-08). Awarded in May, 2009.
 2. **University first rank medal** for securing the highest marks in MCA examinations (Session: 2008-11). Awarded on 5th October, 2013.
 3. **Jindal Jubilee Gold medal** for securing the highest marks in MCA (Session: 2008-11). Awarded on 5th October, 2013.
 4. **Government of West Bengal Merit scholarship** for performance in BCA examinations of 2008. The scholarship was awarded in July 2009.
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Publications

[See *Annexure 1* for publication details]

	International	National
Journals(Indexed)	12	Nil
Conferences	14	01
Book Chapters	04	Nil
Edited Books	02	Nil
Monographs	01	Nil

Other contributory research activities

Member of Editorial Board

1. Scientific Reports, Springer Nature (**H-Index - 250+, Indexed in SCI-E, Scopus**)

Reviewer for Journals

1. IEEE Transactions on Emerging Topics in Computational Intelligence [**SCI, SCOPUS Indexed**]
2. Journal of Systems and Software, ELSEVIER. [**SCI, EI, SCOPUS Indexed**]
3. Journal of Information Security and Applications, ELSEVIER. [**SCOPUS Indexed**]
4. IETE Journal of Research, Taylor and Francis Group. [**SCI, EI, SCOPUS Indexed**]
5. Alexandria Engineering Journal, ELSEVIER. [**SCOPUS Indexed**]
6. Arabian Journal of Science and Engineering [**SCI, SCOPUS Indexed**]
7. Optiks – Journal of Optical Communications, ELSEVIER [**SCOPUS, SCI Indexed**]
8. IET Information Security Journal, ELSEVIER [**SCOPUS, SCI Indexed**]
9. King Saud University Journal of Computer Science, ELSEVIER [**SCOPUS, SCI-E Indexed**]
10. International Journal of Information Management, ELSEVIER [**SCOPUS, SSCI Indexed**]
11. Indonesian Journal of Electrical Engineering and Computer Science [**SCOPUS Indexed**]
12. Journal of Neural Engineering [**SCOPUS, SCI Indexed**]
13. Signal, Image and Video Processing [**SCOPUS, SCI Indexed**]
14. Cluster Computing [**SCOPUS, SCI Indexed**]
15. Scientific Reports [**SCOPUS, SCI Indexed**]
16. Iranian Journal of Science and Technology [**SCOPUS, SCI Indexed**]
17. NPJ Heritage Science Journal [**SCOPUS, SCI Indexed**]

Areas of Interest

Image Steganography and Watermarking, Cryptology, Cyber Security, Information Theory, Machine Intelligence, Cyber Psychology.

Technical Skills

Computer Languages: C, Java(Core),VB/VB.NET, MATLAB, Python, R, Solidity

Software Proficiency: MS-OFFICE, Adobe Photoshop CS5, Microsoft Visual Studio 2008,
SPSS 14.

Operating Systems: Linux, Windows (XP and higher)

Personal Details

Date of Birth: 24-September-1986

Nationality: Indian

Language Proficiency: English (Read and Write), Hindi (Read and Write) and Bengali (Read and Write).

DECLARATION

I hereby declare that all information supplied above is true to the best of my knowledge and that no part of it has been wrongly altered or mutilated.

(Ratnakirti Roy)

List of Publications

International Journals

1. K Ramakrishna Reddy, **Ratnakirti Roy**, Multi-Head Probsparse Spherical Convolutional Network With Lyrebird Optimization Algorithm For Prostate Cancer Identification, Journal of Carcinogenesis, Vol.24, No. 7s, pp. 726-737, 2025 (**Indexed in Scopus Q3**)
2. K Ramakrishna Reddy, **Ratnakirti Roy**, Silent Sentinel: The Unseen Battle of Prostate Cancer, Panamerican Mathematical Journal, Vol.35, No. 4s, pp. 136-148, 2025 (**Indexed in Scopus Q4**)
3. Soma Debnath, **Ratnakirti Roy**, Suvamoy Changder, "Photo Classification Based on the Presence of Diagonal Line using Pre-trained DCNN VGG16", Multimedia Tools and Applications, Vol. 81, No. 16, Springer, 2022 (**Indexed in SCI-E, SCOPUS, Web Science, SCOPUS Q1**)
4. Soma Debnath, **Ratnakirti Roy**, Suvamoy Changder, "A novel approach using deep convolutional neural network to classify the photographs based on leading-line by fine-tuning the pre-trained VGG16 neural network", Multimedia Tools and Applications, Springer, 2022 (Accepted for Publication) (**Indexed in SCI-E, SCOPUS, Web Science, SCOPUS Q1**)
5. Dipjyoti Das, **Ratnakirti Roy**, "Pandemic rise of the Social Media: Echoing fears of social isolation and displacement in the urban Indian context", Humanities and Social Science Studies, Vol. 8, No. 1, 2019, pp. 157-167 (**UGC Approved**)
6. Sanchita Pathak, **Ratnakirti Roy**, Suvamoy Changder, "Performance analysis of image steganalysis techniques and future research directives", International Journal of Information and Computer Security, Inderscience , Vol. 10, No. 1, 2018, pp. 1-24 (**Indexed in SCOPUS, DBLP, UGC Approved, SCOPUS Q3**)
7. **Ratnakirti Roy**, Tauheed Ahmed, Suvamoy Changder, "Watermarking through Image Geometry Change Tracking", Visual Informatics, Vol.2, No. 2, pp. 125-135, Elsevier, 2018 (**Indexed in SCOPUS, Web of Science, UGC Approved, SCOPUS Q1**)
8. **Ratnakirti Roy**, Dipjyoti Das, "Dilution of Social Media Privacy: Effects, Threats and Alternatives", Media Watch Journal, Vol. 8, No. 3, 2017, pp. 401-412 (**Indexed in SCOPUS, UGC Approved**)
9. **Ratnakirti Roy**, Shabnam Samima, Suvamoy Changder, "A Map based Image Steganography scheme for RGB Images", International Journal of Information and Computer Security, Inderscience Publishers, Vol. 7, No. 2/3/4, pp. 196-215, 2015. (**Indexed in SCOPUS, DBLP, UGC Approved, SCOPUS Q3**).
10. **Ratnakirti Roy**, Suvamoy Changder, "Quality Evaluation of Image Steganography Techniques: A Heuristics based Approach", International Journal of Security and its Applications (ISSN: 1738-9976), Vol. 10, No. 4, pp. 179-196, 2016 (**Indexed in SCOPUS, EI, Web of Science**)

11. **Ratnakirti Roy**, Suvamoy Changder, "Steganography with Projection Aided Payload Dimension Reduction and Reconstruction for military covert communication", International Journal of Computer Applications (ISSN: 0975 – 8887), Vol. 139, No. 3, pp. 32-37, **2016**.
12. **Ratnakirti Roy**, K Ramesh Adiga, "A Machine Learning based Approach for Analysis and Prediction of Rice Yield in Semi-Arid Regions of India", Recent Research Reviews, ISSN: 2583-7079, 2025 (Accepted for Publication).

International Conferences

1. Sheela Maharajpet, **Ratnakirti Roy**, Cheeranjeevi S, "The Intersection Of Technology And Driving: A Review Of Advanced Driver Assistance Systems", IEEE ICRISST, Bengaluru, March **2024**, DOI: <https://doi.org/10.1109/ICRISST59181.2024.10922025>. (**SCOPUS Indexed**)
2. Krishna Roy, Ayan Mukherjee, **Ratnakirti Roy**, "Relation between Time of the News Publishing and Probability of Fake News", First International Conference on Strategic HR Under Technological Innovation (SHRUTI 2022), Calcutta Business School, **2022**.
3. Arunava Mookherjee, Krishna Roy, Sujata Banerjee, **Ratnakirti Roy**, "Should Schools Open Physically - An Empirical Study", Proc. ISDSI-Global, December 27-30, **2021**, IIM Nagpur, India.
4. Ayan Mukherjee, **Ratnakirti Roy**, "A Deep Learning Approach towards Determining the Effects of News Trust Factor Based On Source Polarity", Proc. ICTCS - 2020, December 11-12, **2020**, Jaipur, India.
5. Agnihotri Roy, Sumon Dey, **Ratnakirti Roy**, "Big data driven predictive mathematical model for improving students' performance in undergraduate level examinations in India", International Conference on Advances in Computational Mathematics, Tribhuvan University, Kathmandu, Nepal, December 23-24, **2018**.
6. Sabyasachi Kamila, **Ratnakirti Roy**, Suvamoy Changder, "A DWT based Steganography Scheme with Image Block Partitioning", Proc. SPIN 2015, Amity School of Engineering and Technology, Amity University, Noida, UP, February 19-20, **2015**, pp. 471-476. (**SCOPUS Indexed**)
7. Avinash Tyagi, **Ratnakirti Roy**, Suvamoy Changder, "High Capacity Image Steganography based on Pixel Value Differencing and Pixel Value Sum", 2nd International Conference on Advances in Computing and Communication Engineering, Dehradun, 1-2 May, **2015**, pp. 488-493. (**SCOPUS Indexed**)
8. **Ratnakirti Roy**, Suvamoy Changder, "Image Steganography with Block Entropy based Segmentation and Variable Rate Embedding", 2nd International Conference on Business and Information Management (ICBIM 2014) [IEEE], National Institute of Technology, Durgapur, pp. 75-80, **2014**. (**SCOPUS Indexed**)

9. **Ratnakirti Roy**, Suvamoy Changder, "Image Realization Steganography with LCS based Mapping", Proc. 2014 7th International Conference on Contemporary Computing (IC3 2014), IIIT, Noida, 7-9 August, 2014, pp. 218-223. [Proceedings available on IEEEExplore] (**SCOPUS Indexed**)
10. Piyush Kumar Gupta, **Ratnakirti Roy**, Suvamoy Changder, "A Secure Image Steganography Technique with Moderately Higher Significant Bit Embedding", 2014 International Conference on Computer Communication and Informatics [IEEE], Coimbatore, January 3-5, 2014, pp. 1-6. (**SCOPUS Indexed**)
11. **Ratnakirti Roy**, Suvamoy Changder, Anirban Sarkar, Narayan C Debnath, "Evaluating image steganography techniques: Future research challenges", 2013 International Conference on Computing, Management and Telecommunications (ComManTel) [IEEE], pp.309-314, 21-24 Jan. 2013, DOI: 10.1109/ComManTel.2013.6482411. (**SCOPUS Indexed**)
12. **Ratnakirti Roy**, Suvamoy Changder, Anirban Sarkar, "SKINHIDE: A Biometric Inspired High Fidelity Steganography Technique", Proc. 3rd International Conference on Advanced Computing and Communications, Los Angeles, USA, pp. 41-46, 2013. ISBN: 978-1-880843-91-8.
13. **Ratnakirti Roy**, Anirban Sarkar, Suvamoy Changder, "Chaos based Edge Adaptive Image Steganography", Procedia Technology, Elsevier, Vol. 10, No.1, 2013, pp. 138-146. DOI: 10.1016/j.protcy.2013.12.346.
14. Shabnam Samima, **Ratnakirti Roy**, Suvamoy Changder, "Secure Key based Image Realization Steganography", IEEE International Conference on Image Information Processing, Jaypee University of Information Technology, Shimla, pp. 377-382, 2013. (**SCOPUS Indexed**)

National Conferences

1. Sumon Dey, **Ratnakirti Roy**, "Autofocus technologies in digital cameras : A comparative study", Interdisciplinary Conference in Future Paradigms in Industry, Inspiria Knowledge Campus, Siliguri, November 1-2, 2018.

Book Chapters

1. Sumit Singha Chowdhury, **Ratnakirti Roy**, "Aqua Kit: A Kit for Efficient Freshwater Fish Farming – A Comprehensive Sensor-Based Approach", Leading the Charge: A Guide to Management, Entrepreneurship and Technology in the Dynamic Business Landscape Edition 1, 14 January 2025, Page 206-217, <https://doi.org/10.9734/bpi/mono/978-93-48859-98-3/CH13>
2. Suvamoy Changder, **Ratnakirti Roy**, Narayan C. Debnath, "Data Security and Privacy through Text Steganography Approaches", Research and practices on Security Paradigms and Applications of IoT, CRC Press, Taylor and Francis Group, 2020, pp. 193-216. (**SCOPUS Indexed**)

3. **Ratnakirti Roy**, Suvamoy Changder, "Object based Image Steganography with pixel polygon tracing", Computational Intelligence in Data Mining - Volume 1, Smart Innovation, Systems and Technologies Volume 31, 2015, pp 699-709, Springer India. **(SCOPUS Indexed)**
4. Tauheed Ahmed, **Ratnakirti Roy**, Suvamoy Changder, "Cropping and Rotation Invariant Watermarking Scheme in the Spatial Domain", Advances in Intelligent Systems and Computing, Vol. 249, pp. 281-292, Springer-Verlag, Berlin, 2013. **(SCOPUS Indexed)**

Books & Monographs

1. **Ratnakirti Roy, Sumit Singha Chowdhury, Hanamant Jakaraddi (Eds.)**, "Intelligent IoT Systems: From Research to Real-World Solutions", BP International Publications, 2025, ISBN - 978-93-88417-23-5.
2. **Ratnakirti Roy, Juin Choudhury, Monica M (Eds.)**, "Leading the Charge: A Guide to Management, Entrepreneurship and Technology in the Dynamic Business Landscape", BP International Publications, 2025, ISBN-978-93-48859-98-3
3. **Ratnakirti Roy, Manish Thakur, Anushree Raj (Eds.)**, "Convergence of Machine Learning and IoT for Enabling the Future of Intelligent Systems", QTAalytics, 2024, ISBN-978-81-966500-7-0
4. **Ratnakirti Roy**, "Object based Steganography - Novel Approaches to Secure and Robust Data Hiding", Lambert Academic Publishers, 2023, ISBN-978-620-6-79011-2.